

Quiz: Portfolio Optimization

1 Scenario analysis of the holding period return

Shares in Kojo currently trade for NOK 100, and the company's dividend policy is to pay out 8% of its retained earnings as dividends at the end of each year. In the course of your analysis as an analyst at Kingdom Investments, you determine that the earnings and price of Kojo's shares depend on the state of the economy, and you produce the expectations about Kojo's stock price and dividend shown below.

	Probability	Ending price per share (NOK)	Dividend per share (NOK)
Boom	0.35	140	10.4
Normal growth	0.30	110	8.0
Recession	0.35	80	5.6

Based on these expectations, compute the mean and standard deviation of the holding period return (HPR) on this stock.

2 Arithmetic and geometric average returns

A share in Fund C has had the following value developments over the last three years:

$$t_0 = \text{NOK } 200; \quad t_1 = \text{NOK } 224; \quad t_2 = \text{NOK } 280; \quad t_3 = \text{NOK } 238$$

- Calculate the arithmetic average return over the three years.
- Calculate the geometric average return over the three years.
- Briefly explain why the two methods in (a) and (b) provide different answers.

3 Using historical returns to estimate expected returns

Your supervisor has just given you a dataset of the annual rates of return on Norwegian stocks all the way back to 2000. Is there any advantage or disadvantage of using this dataset to estimate the expected return on Norwegian stocks over the coming year?

4 The complete portfolio and the optimal risky share

Damaris Johnson manages a risky portfolio with an expected return of 18% and a standard deviation of 28%. The T-bill rate is 8%.

- a) Suppose that a new client of Johnson decides to invest a certain portion of their investment in the risky portfolio (y) and the remainder in T-bills ($1 - y$), so that the overall portfolio will have an expected rate of return of 16%.
 - i. What proportion is invested in the risky portfolio?
 - ii. What is the standard deviation of this portfolio?
- b) Suppose that the new client prefers to invest an amount in the risky portfolio that maximises the expected return of the complete portfolio, subject to the constraint that the standard deviation of the complete portfolio cannot be greater than 18%.
 - i. What proportion is invested in the risky portfolio?
 - ii. What is the expected return of the complete portfolio?
- c) Through an analysis of the client's financial profile, it is determined that their degree of risk aversion is $A = 5$.
 - i. What proportion y should now be invested in the risky portfolio?
 - ii. What are the expected return and standard deviation of the client's optimised portfolio?

5 The maximum risk aversion that still prefers the risky portfolio

Consider a portfolio that offers an expected rate of return of 12% and a standard deviation of 18%. T-bills offer a risk-free 7% rate of return. What is the maximum level of risk aversion for which the risky portfolio is still preferred to T-bills?

Note: $U = E(r) - 0.5 A \sigma^2$.

6 Portfolio of two stocks

Suppose that there are only two stocks in the market: stock Alfa and stock Bravo. The expected return on Alfa is 15% with a standard deviation of 10%. Bravo has an expected return of 12% with a standard deviation of 7%. The covariance between the returns of the stocks is 0.002 (in decimals). The annual risk-free rate is 3%. Assume that short selling is not allowed.

	Stock Alfa	Stock Bravo
Expected return, μ	15%	12%
Standard deviation, σ	10%	7%
Covariance, σ_{AB}	0.002	

- What is the minimum standard deviation you can obtain by investing in a combination of Alfa and Bravo?
- What would your asset allocation be if you want to maximize the risk-adjusted return (Sharpe ratio) of the portfolio? What is the Sharpe ratio of this portfolio?

7 Expected return and risk of a client's complete portfolio

You manage an equity fund with an expected risk premium of 10% and an expected standard deviation of 14%. The rate on Treasury bills is 6%. Your client chooses to invest NOK 60,000 of her portfolio in your equity fund and NOK 40,000 in a T-bill money market fund. What are the expected return and standard deviation of return on your client's portfolio?

8 The expected risk premium in kroner

Given NOK 100,000 to invest, what is the expected risk premium in NOK of investing in equities versus risk-free T-bills, based on the information below?

Action	Probability	Expected return
Invest in equities	0.6	NOK 50,000
	0.4	−NOK 20,000
Invest in risk-free T-bill	1.0	NOK 5,000

9 Choosing between funds and combining equity with bonds

Assume that an investor's preferences can be expressed by the utility function $U = E(r) - 0.005 \cdot A \cdot \sigma^2$, where U is utility, $E(r)$ is the expected return, A is the risk aversion coefficient and σ^2 is the variance.

A risk-averse investor has invested NOK 1 million in Fund A, which offers an expected return of 15% and a standard deviation of 25%. A financial advisor has offered Fund V, which offers an expected rate of return of 16% and a standard deviation of 28%, as an alternative.

- a) If the investor has a risk aversion coefficient of 3, which is the better of Funds A and V?
- b) For which risk aversion coefficient will the two funds be equally good?

Assume now that there exists a bond investment, Fund B, with an expected return of 8% and a standard deviation of 20%. The correlation coefficient between the return on Fund A and Fund B is -0.2 . This investor is considering combining the equity and bond funds in a portfolio P.

- c) What share invested in Fund A minimises the risk of portfolio P?
- d) Compute the expected return and standard deviation of portfolio P.